THE DEPTH OF A COHEN-MACAULAY MODULE AGAINST AN IDEAL AS A DIFFERENCE OF DIMENSIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of the associated primes of a module when the depth against an ideal is zero. Invoked inside the counted proof (source lines 2073-2075). Places the ideal inside an associated prime in the zero depth case.
- The equality of the dimension of the support with the dimension of the quotient by a minimal associated prime for a Cohen-Macaulay module. Invoked inside the counted proof (source lines 2077-2079). Turns the inequality in the zero depth case into an equality.
- The drop of the depth by exactly one after dividing by a nonzerodivisor in the ideal. Invoked inside the counted proof (source lines 2086-2087). Provides the induction step on the depth.
- The drop of the dimension of the support by exactly one after dividing by a nonzerodivisor. Invoked inside the counted proof (source lines 2088-2089). Keeps the two sides of the identity in step during the induction.

Source reference. Original reference algebra-lemma-CM-ass-minimal-support: sources/stacks-dualizing/pinned-original-source.tex, source lines 2079–2079. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-CM-ass-minimal-support, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference algebra-lemma-ideal-nonzerodivisor: sources/stacks-dualizing/pinned-original-source.tex, source lines 2075–2075. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-ideal-nonzerodivisor, which belongs to another chapter of the project and is not part of this card. The

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Source reference. Original reference algebra-lemma-one-equation-module: sources/stacks-dualizing/pinned-original-source.tex, source lines 2089–2089. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-one-equation-module, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-depth-drops-by-one: sources/stacks-dualizing/pinned-original-source.tex, source lines 2039–2039. Navigation pointer only: the counted text refers to the Stacks item lemma-depth-drops-by-one, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let R be a Noetherian local ring. If M is a finite Cohen-Macaulay R-module and $I \subset R$ a nontrivial ideal. Then*

$$\mathit{depth}_I(M) = \dim(\mathit{Supp}(M)) - \dim(\mathit{Supp}(M/IM)).$$

Proof. We will prove this by induction on $\text{depth}_I(M)$.

If $\text{depth}_I(M) = 0$, then I is contained in one of the associated primes $\mathfrak{p}$ of M (Algebra, Lemma algebra-lemma-ideal-nonzerodivisor). Then $\mathfrak{p} \in \text{Supp}(M/IM)$, hence $\dim(\text{Supp}(M/IM)) \geq \dim(R/\mathfrak{p}) = \dim(\text{Supp}(M))$ where equality holds by Algebra, Lemma algebra-lemma-CM-ass-minimal-support. Thus the lemma holds in this case.

If $\text{depth}_I(M) > 0$, we pick $x \in I$ which is a nonzerodivisor on M. Note that $(M/xM)/I(M/xM) = M/IM$. On the other hand we have $\text{depth}_I(M/xM) = \text{depth}_I(M) - 1$ by Lemma lemma-depth-drops-by-one and $\dim(\text{Supp}(M/xM)) = \dim(\text{Supp}(M)) - 1$ by Algebra, Lemma algebra-lemma-one-equation-module. Thus the result by induction hypothesis. □